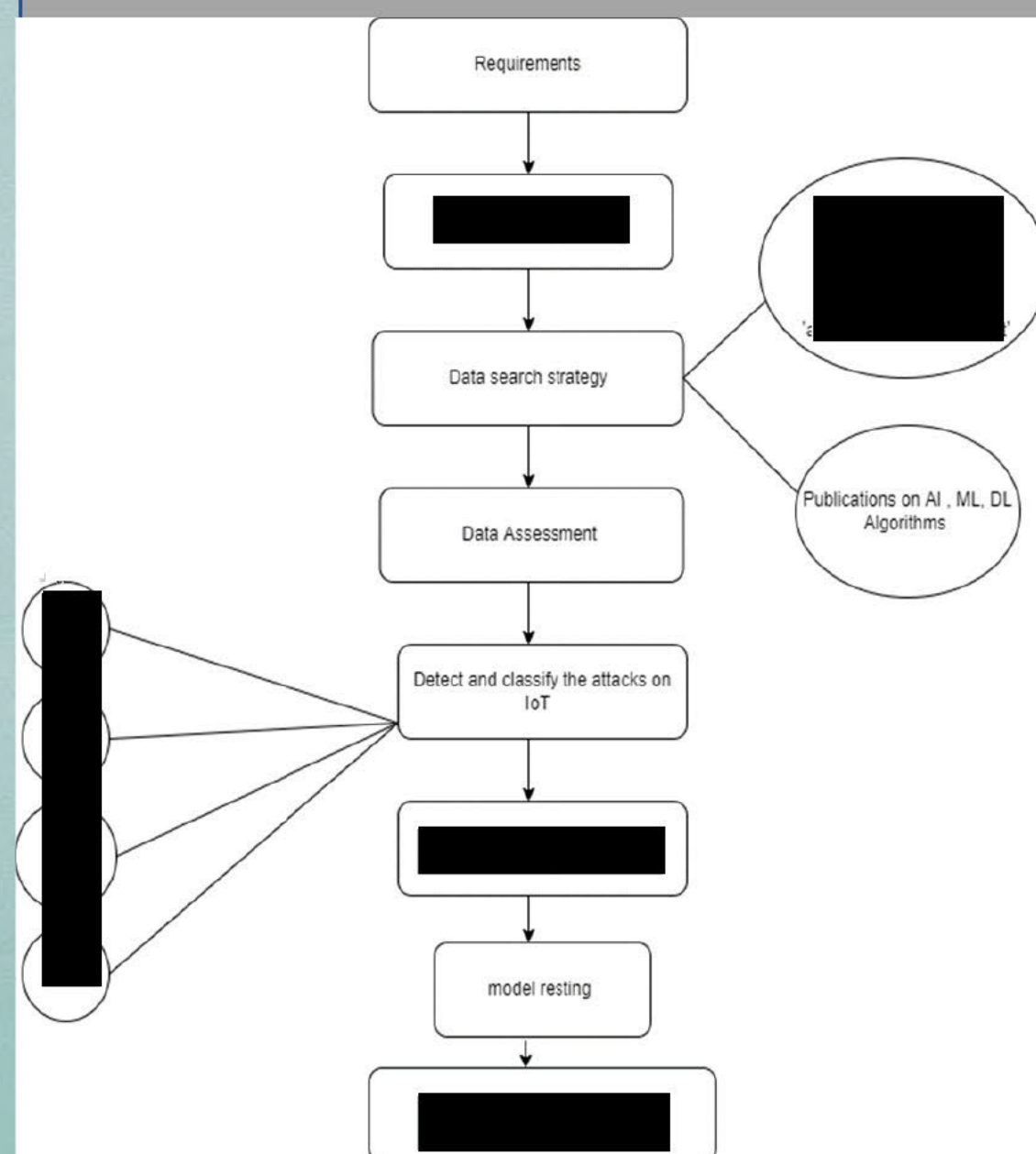


ABSTRACT: The rapid proliferation of IoT devices and the increasing reliance on cloud services have led to increased cybersecurity challenges. This paper explores the integration of Artificial Intelligence (AI) and Machine Learning (ML) into existing security frameworks to enhance threat detection and response. The study reviews current security approaches for IoT environments and provides a systematic literature review of AI and ML applications in cybersecurity. Key trends in cybersecurity research are identified, and the potential of AI and ML to address these challenges is examined. The results suggest that AI and ML can significantly improve security posture, but their implementation requires careful consideration of privacy, security, and ethical implications. Future research should focus on developing robust, scalable, and explainable AI/ML models for cybersecurity.

WORKFLOW



METHODOLOGY: The can be used to detect prob trees: These models use a t series of decisions and t Decision trees can be used either normal or abnorm source IP address, desti number. Random forests: T of decision trees. Random the accuracy of decisi predictions of multiple tree used to classify network tra finding the best boundary the normal data from t Networks: These models nodes, called neurons, t predictions. Neural netw network traffic as normal such as source IP address, c number. KNN: This algo network traffic as normal c features of a new observati seen ob

CONCLUSION: account, from all the have collected and i able to present ou cybersecurity attac attacks are clas interception, attac result in faulty fee witnessed that the intelligence (AI) meth has grown in popul the help of all the info have assembled fr publications, we w better in-depth anal provide the most suc detecting attacks in analyze the availab those attacks utilizi through

Introduction : AI is a computer science that aims to replicate the capabilities of the human mind. It is being invested in by technology across various industries. Currently, there is no acknowledged definition. It explains that the goal is to recreate or reproduce human intelligence in machines. The traditional method for detecting intrusions is becoming increasingly outdated. That AI and machine learning are becoming increasingly important in information security. These technologies are able to analyze large amounts of data to identify a wide range of threats, including malware and phishing attempts. Currently, AI and machine learning continuously adapt, helping to identify new types of threats in the present and future.

Research Objective:

is to assess the viability of a two-layer neural network. When a hidden layer is added to a two-layer network, the back-propagation becomes more difficult and time-consuming. The resulting network is less memory-efficient. As a result, it is always preferable to use a two-layer classifier to determine whether it will produce the same outcomes as a more complex structure that is less computationally effective. That is simpler. Additionally, the process will

RESULT

Table VI. Performance Comparison of Different Classification Algorithms

Classifier	Metric	Type of Attack			
		Buffer_Overflow	loadmodule	Perl	rootkit
SVM	Accuracy		93.89		
	Precision	97.09	97.86	75.54	90.33
	Recall	93.91	93.53	95.86	93.97
SVM	Accuracy		91.73		
	Precision	85.74	93.22	84.09	100
	Recall	100	99.89	51.03	77.87
SVM	Accuracy		98.11		
	Precision	98.35	97.56	96.12	100
	Recall	97.66	100	85.52	99.14
SVM	Accuracy		99.50		
	Precision	99.76	99.77	96	100
	Recall	98.59	100	99.31	99.43
SVM	Accuracy		85.06		
	Precision	80.19	84.29	100	100
	Recall	99.53	99.89	2.07	64.37
SVM	Accuracy		92.06		
	Precision	75.18	99.77	100	100
	Recall	100	96.82	22.07	99.43
SVM	Accuracy		91.8		
	Precision	76.94	98.52	83.33	100
	Recall	100	98.18	20.69	95.40
SVM	Accuracy		89.67		
	Precision	81.84	96.47	41.67	93.37
	Recall	93.91	96.03	20.69	97.13
SVM	Accuracy		96.78		
	Precision	96.89	98.97	99.22	99.77
	Recall	9.85	98.41	87.39	98.85
SVM	Accuracy		99.44		
	Precision	99.53	99.77	95.92	100
	Recall	98.59	100	97.24	100
SVM	Accuracy		99.06		
	Precision	99.53	99.21	94.44	100
	Recall	98.59	99.77	93.79	100

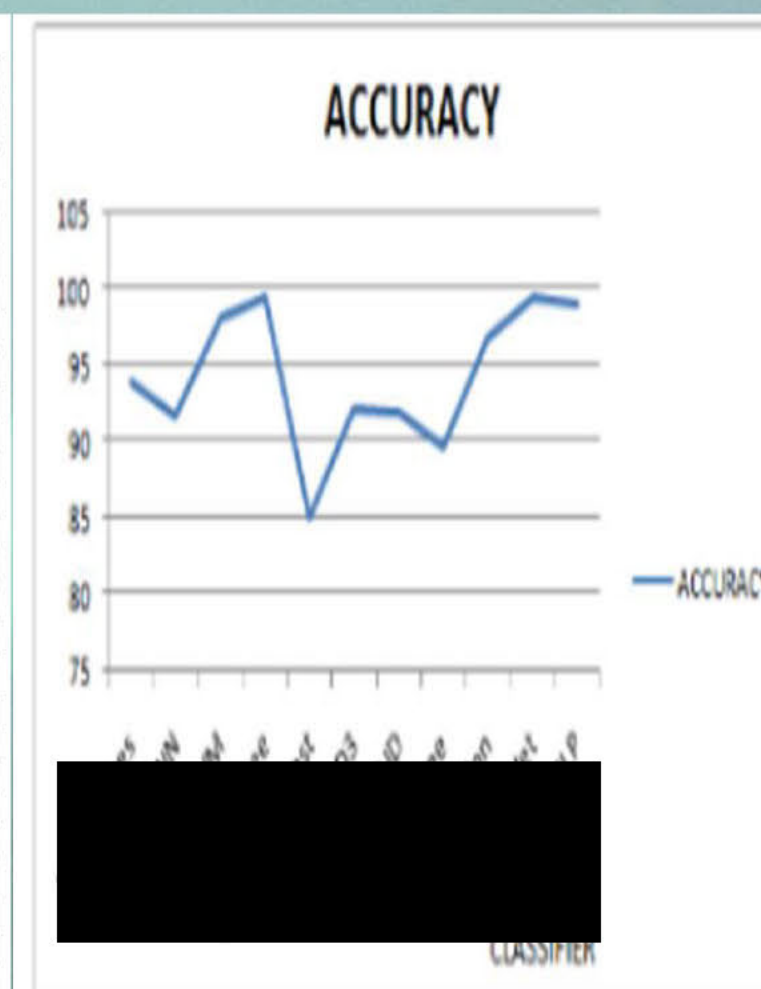


Fig2.  algorithms

Ref <https://>

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